Entity-Relationship (ER) Conceptual Architecture

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1. Conceptual Model Summary (22 Relational Tables)

The Intelligence Fusion Platform features a **22-table relational schema** structured into **10 primary entity/audit relations** and **12 associative junction tables** resolving all Many-to-Many entity relationships in 3NF and BCNF, linked to a flexible MongoDB document repository.

Entity	Type	Primary Key	Key Relationships & Cardinality
officers	Core Personnel	officer_id (PK)	1:N to cases (leads), 1:N to evidence (seizes), 1:N to custody
cases	Master Incident	case_id (PK)	1:N to evidence, 1:N to case_activity, M:N to all entities via bridges
people	Subject / Witness	person_id (PK)	M:N to cases, M:N to organizations, M:N to vehicles, M:N to devices
organizations	Corporate / Syndicate	organization_id (PK)	M:N to cases, M:N to people
vehicles	Transport Exhibit	vehicle_id (PK)	M:N to cases, M:N to people, M:N to evidence
devices	Hardware Exhibit	device_id (PK)	M:N to cases, M:N to people, M:N to evidence
locations	Crime Scene / GPS	location_id (PK)	M:N to cases, M:N to evidence
evidence	Legal Exhibit	evidence_id (PK)	N:1 to cases, 1:N to evidence_custody, M:N to recovery scenes
evidence_custody	Immutable Audit	custody_event_id (PK)	N:1 to evidence (CASCADE), N:1 to officers (RESTRICT)
case_activity	Database Audit Log	activity_id (PK)	N:1 to cases (CASCADE), N:1 to officers (SET NULL)

2. 12 Associative Junction Tables (M:N Normalization)

Junction Table	Composite PK	Foreign Keys	Attributes
case_people	(case_id, person_id)	cases(case_id), people(person_id)	role_in_case, notes, added_at
case_organizations	(case_id, organization_id)	cases(case_id), organizations(organization_id)	association_type, notes, added_at
case_vehicles	(case_id, vehicle_id)	cases(case_id), vehicles(vehicle_id)	role_in_case, notes, added_at
case_devices	(case_id, device_id)	cases(case_id), devices(device_id)	role_in_case, notes, added_at
case_locations	(case_id, location_id)	cases(case_id), locations(location_id)	association_type, notes, added_at
person_organizations	(person_id, organization_id)	people(person_id), organizations(organization_id)	role, added_at
person_vehicles	(person_id, vehicle_id)	people(person_id), vehicles(vehicle_id)	relationship_type, added_at
person_devices	(person_id, device_id)	people(person_id), devices(device_id)	relationship_type, added_at
evidence_people	(evidence_id, person_id)	evidence(evidence_id), people(person_id)	association_type, added_at
evidence_vehicles	(evidence_id, vehicle_id)	evidence(evidence_id), vehicles(vehicle_id)	association_type, added_at
evidence_devices	(evidence_id, device_id)	evidence(evidence_id), devices(device_id)	association_type, added_at
evidence_locations	(evidence_id, location_id)	evidence(evidence_id), locations(location_id)	association_type, added_at